# Boiler



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

**Covers these lab items:** Boiler

## What it is used for

A boiler heats water or another approved fluid to provide hot water, steam, or process heat for kitchen, food-processing, or other industrial applications. Operate it only within its rated pressure, temperature, and fluid limits, and follow the approved startup, monitoring, and shutdown procedures to prevent scalding, pressure-related failures, and equipment damage.

## Main parts

- Boiler vessel or heating chamber
- Burner, heating element, or heat source
- Feedwater inlet and level control
- Pressure gauge and temperature indicator
- Safety relief valve and pressure controls
- Blowdown or drain valve, piping, and control panel

## Before you start

1. Confirm the boiler type, rated pressure, operating temperature, and intended process use.
2. Inspect the vessel, piping, valves, gauges, insulation, burner or heating element, and visible connections for damage or leaks.
3. Verify that the water level, feedwater supply, and water quality meet the manufacturer's requirements.
4. Check that the safety relief valve, pressure controls, alarms, and low-water protection are available and not obstructed.
5. Ensure the exhaust or ventilation system and required protective guards are in place.
6. Confirm that only trained, authorised operators are present and that the operating log and emergency procedure are available.

## How to use

1. Review the previous operating log and confirm that no unresolved fault or maintenance restriction remains.
2. Check the feedwater supply, required fluid level, fuel or electrical supply, and control settings.
3. Confirm that all required valves are in their designated startup positions and that vents and drains are configured according to the approved procedure.
4. Start the boiler using the manufacturer's prescribed sequence and monitor the initial heating stage.

5. Observe pressure, temperature, water level, burner status, and alarms as the boiler approaches operating conditions.
6. Increase load gradually and remain within the rated operating limits and approved process requirements.
7. Check accessible piping and connections for leakage, abnormal vibration, unusual noise, or steam release.
8. Maintain the required water level and follow the specified water-treatment and blowdown schedule.
9. If a low-water condition, pressure excursion, abnormal noise, leak, or alarm occurs, follow the emergency procedure and shut down when required.
10. For normal shutdown, reduce the load and stop the heat source using the approved sequence.
11. Allow pressure and temperature to fall to the specified safe levels before opening drains, access covers, or connected equipment.
12. Record operating readings, water treatment, blowdown, alarms, inspections, and maintenance actions in the log.

## ⚠ Safety

- Only qualified and authorised boiler operators should start, operate, or service the equipment.
- Never exceed the boiler's rated pressure or temperature and never bypass safety interlocks or the relief valve.
- Never operate a boiler with inadequate water level, defective pressure controls, or a blocked relief path.
- Treat steam, hot water, surfaces, and piping as burn hazards; use suitable PPE and keep unauthorised people away.
- Do not open pressurised fittings, drains, or access covers; confirm safe pressure and temperature before opening as specified by the procedure.
- If a serious leak, pressure-control failure, or suspected vessel damage occurs, keep clear, isolate the unit only if safe, and contact qualified personnel or emergency responders.

## Care & storage

- Inspect the vessel, burner or heating element, gauges, valves, piping, and insulation according to the maintenance schedule.
- Test safety devices, alarms, and low-water protection at the specified intervals using approved procedures.
- Maintain feedwater quality and water treatment to limit scale, corrosion, and deposits.
- Perform blowdown only at the prescribed intervals and according to the approved procedure.
- Arrange qualified inspection, cleaning, and servicing; keep records of tests, repairs, and operating hours.

## Common mistakes to avoid

- Starting the boiler without checking water level or safety controls.
- Operating beyond rated pressure or temperature.

- Ignoring leaks, unusual noises, alarms, or unstable water level.
- Opening a drain or access cover before pressure and temperature have reached safe levels.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*